

ERVA ŞENGÜL

Computer Engineer

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Computer engineer specializing in artificial intelligence and cloud native systems. Focused on applying AI to real business problems.

EDUCATION

B.S., Computer Engineering | Istanbul Medipol University

Istanbul, Türkiye | Oct 2022 - Jun 2026

Graduation project: Intelligent Cluster Optimizer - a Go-based Kubernetes optimizer combining a statistical engine, ML-based resource forecasting, and a four-layer safety framework. GPA 3.1/4.0

High School Diploma | Istanbul Koy Hizmetleri Anatolian High School

Istanbul, Türkiye | Oct 2018 - Jun 2022

Advanced Physics, Biology and Chemistry route. GPA 93.7/100

PUBLICATIONS

A. Karakaya, E. Sengul, and A. Kaplan. "Safety-Gated Autoscaling: A Multi-Layered Defense Architecture for Kubernetes Vertical Resource Optimization." arXiv:2607.26503, 2026. arxiv.org/abs/2607.26503

EXPERIENCE

Software Engineer Intern | Medipol Healthcare Group

Istanbul, Türkiye | Aug 2025 - Oct 2025

Supported IPAM modules used in international patient operations and hospital information workflows.
Configured, troubleshooted, and supported healthcare IT systems handling secure patient data with high operational reliability.
Coordinated infrastructure and system support in a production healthcare environment with strict data handling requirements.

Engineer Participant | TEKNOFEST 2025

Istanbul, Türkiye | Sep 2025

Presented a real-time emotion recognition application built with Python, TensorFlow, Keras, and OpenCV.
Developed a CNN pipeline for seven-class facial expression recognition using CK+ data augmentation and live webcam inference.

Student Participant | TEI - TUSAS Engine Industries, Inc.

Istanbul, Türkiye | Mar 2025 - May 2025

Completed TEI's competitive professional development program focused on aerospace systems and turbine-engine engineering.
Analyzed engineering case studies, strengthening systems thinking across reliability, performance, and technical problem solving.

Software Engineer Intern | Conovel Information Technologies

Istanbul, Türkiye | Sep 2024 - Oct 2024

Built full-stack features in ASP.NET Core MVC using C#, HTML, CSS, and SQL.
Implemented dynamic components and database-backed functionality to improve usability and system performance.

PROJECTS

Intelligent Cluster Optimizer | Graduation Project - Published (arXiv:2607.26503)

Built a Go-based Kubernetes optimizer that achieved 20-40% cost savings on Google Kubernetes Engine through ML-driven vertical right-sizing.
Designed a statistical engine, Holt-Winters forecasting, and a four-layer safety framework with memory-leak gating; validated with 1118 automated tests at 80.3% coverage.

AI Pre-Evaluation System for Medical Tourism

Fine-tuned MedGemma-4B (QLoRA) to convert multilingual clinical documents into structured JSON; built an NLLB-200 translation pipeline with overlapping-window chunking and a deterministic clinical timeline.

Real-Time Emotion Recognition (TEKNOFEST 2025)

Built a real-time computer-vision app classifying seven facial emotions from webcam input using Python, TensorFlow, Keras, and OpenCV.

Weather Forecast Modeling Using Interpolation

Built a forecasting workflow on NOAA data with Python preprocessing, interpolation, and time-series analysis for seasonal pattern detection.

PIC16F877A-Based Car Battery Voltage Monitoring System

Integrated circuit design, firmware programming, and hardware debugging for an embedded automotive monitoring prototype using PICkit 3.

TECHNICAL SKILLS

Languages: Python, Go, C#, C, SQL, JavaScript, HTML/CSS

AI / ML: Claude, Codex, Gemini, Copilot, TensorFlow, Keras, PyTorch, OpenCV, scikit-learn, Hugging Face, NLP, Computer Vision

Cloud & DevOps: Kubernetes, GKE, Docker, Git, Prometheus